4.3

Sustainable Production of Goods and Services

Cambridge IGCSE Business

- What is Sustainability?
- Renewable Energy
- Reducing Waste
- Reuse & Recycle
- Eco Products & Packaging
- Advantages & Disadvantages

Sustainable Production

Sustainable production means meeting today's needs without compromising the ability of future generations to meet their own needs — the UN Brundtland definition.

- Use renewable energy sources
- Reduce, reuse and recycle materials
- Develop environmentally friendly products
- Minimise carbon footprint and pollution
- Evaluate the business case for sustainability

What is Sustainable Production?

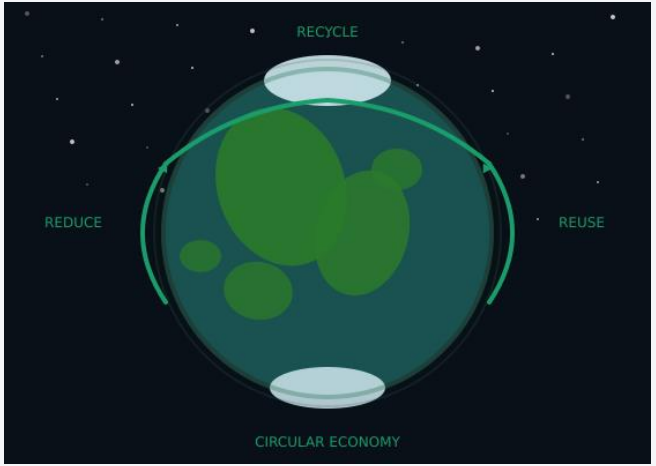
Meeting today's needs without harming the future

Sustainable Production – Definition & Why It Matters

Sustainable production means making goods and services in ways that minimise environmental damage, conserve natural resources and reduce pollution for future generations.

Why businesses are adopting sustainable production:

- Customer pressure — 73% of global consumers say they would change habits for the environment
- Legal requirements — governments increasingly mandate emissions limits, waste targets, packaging rules
- Cost savings — renewable energy and waste reduction lower long-term operating costs significantly
- Brand reputation — sustainable businesses attract talent, investors and eco-conscious customers
- Climate change urgency — businesses face real consequences from floods, droughts and resource scarcity



Leading sustainable businesses:

Patagonia
1% of sales to environment
Repair over replace
★ Leader in eco-fashion

IKEA
1m+ solar panels on stores
100% renewable energy
★ Net positive by 2030 target

Unilever
CO_2 cut 70% since 2008
Sustainable Living Plan
★ 400+ sustainable brands

Patagonia · IKEA · Unilever · Adidas · Coca-Cola · Tesla

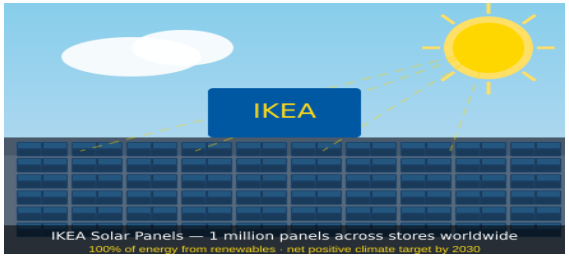
Methods of Sustainable Production

Renewable energy · Less waste · Reuse · Recycle · Eco products

Sustainable Production – Six Key Methods with Real Examples

There are six main ways businesses can become more sustainable. Most successful businesses use a combination of all six.

Renewable Energy

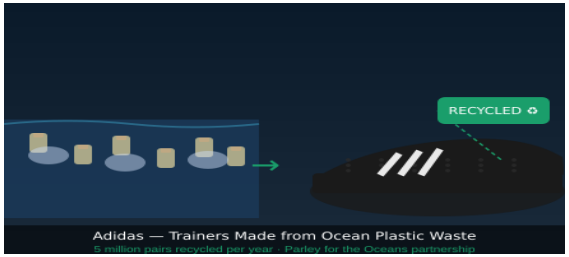


IKEA: 1m solar panels · 100% renewable energy

Reduce Waste

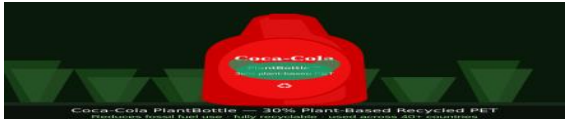
Unilever: 97% less waste since 2008

Recycle



Adidas: 5m pairs/yr from ocean plastic

Reuse Resources



Coca-Cola PlantBottle: 30% plant-based PET · recyclable

Eco-Friendly Products

Patagonia: recycled fleece · free Worn Wear repairs

Eco-Friendly Packaging

McDonald's: 1.8bn plastic straws eliminated per year

Sustainable Production – Renewable Energy in Detail

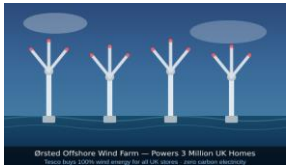
Switching to renewable energy is the most visible sustainable change a business can make. It reduces carbon emissions and long-term energy costs.

Solar Power

Panels on roofs or solar farms convert sunlight directly into electricity.

IKEA installed 1 million solar panels on its stores. Apple's HQ runs on 100% solar.

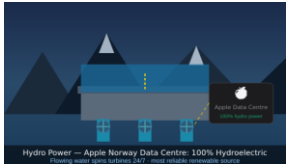
Wind Power



Wind turbines generate electricity — offshore turbines are very efficient.

Ørsted wind farms power 3m UK homes. Tesco buys wind energy for all its stores.

Hydro Power



Flowing water spins turbines — very reliable and generates continuous power.

Apple's data centres in Norway run on 100% hydroelectric power.

Heat Pumps

Extract heat from ground or air — much more efficient than gas boilers.

Marks & Spencer retrofitting heat pumps into all stores to cut gas use.

Advantages & Disadvantages

The business case for sustainable production

Sustainable Production – Advantages & Disadvantages

Becoming more sustainable is not just good for the planet — it can also make good business sense. But there are real costs and challenges involved.

Advantages

- Improved reputation — attracts environmentally conscious customers
- Cost savings — renewable energy and waste reduction lower bills
- Meets legal requirements — avoids fines and regulatory penalties
- Attracts investors — ESG funds now worth trillions globally
- Motivates employees — staff want to work for ethical companies
- Competitive advantage — differentiates from less sustainable rivals

Disadvantages

- High upfront costs — solar panels, wind turbines, new machinery
- Takes time to see financial returns on the investment
- Supply chain changes are complex and disruptive
- Sustainable materials can be more expensive to source
- Consumers don't always pay a premium for green products
- Risk of greenwashing accusations — claims must be genuine

Business case examples:

Greenwashing – When Sustainability Claims Are False

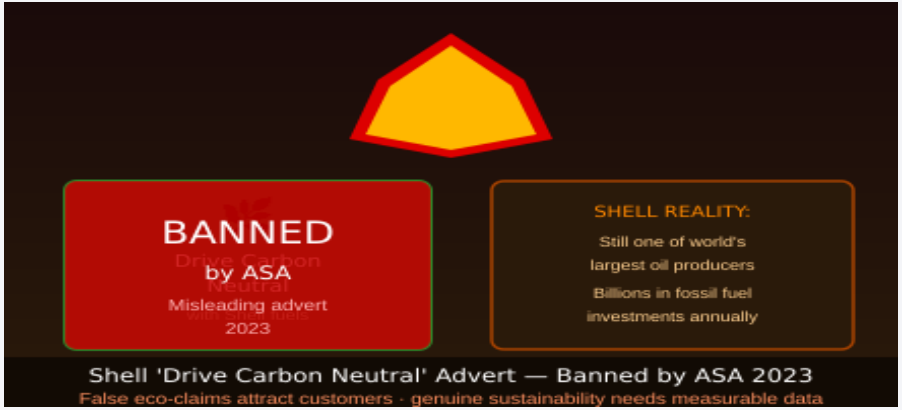
Greenwashing is when a business makes misleading or exaggerated claims about its environmental credentials to attract customers without genuinely being sustainable.

Warning signs of greenwashing:

- Vague claims: 'eco-friendly' or 'natural' with no evidence
- Hidden trade-offs: 'recycled' packaging but product causes pollution
- Irrelevant claims: 'CFC-free' when CFCs were already banned by law
- False imagery: forests and animals used to imply sustainability
- No proof: no certified data, third-party audits or measurable targets

Real greenwashing cases:

- Volkswagen Dieselgate
- Shell “Drive Carbon Neutral” Advert



Also: H&M Conscious Collection investigated by EU (2023) for misleading eco-labelling on fast fashion. Genuine sustainability must be backed by

Key Revision Checklist – 4.3 Sustainable Production of Goods and Services

Definition	Meeting today's needs without compromising future generations. Driven by customer pressure, legal requirements, cost savings and brand reputation
Renewable Energy	Solar (IKEA), wind (Tesco/Ørsted), hydro (Apple). Reduces CO_2 and long-term energy costs — high upfront investment
Reduce Waste	Lean production, less packaging, energy efficiency. Example: Unilever cut manufacturing waste by 97% in 12 years
Reuse	Reuse packaging, components or materials. Extends product life. Example: Coca-Cola PlantBottle — 30% plant-based PET
Recycle	Collect and reprocess materials. Example: Adidas makes trainers from ocean plastic — 5 million pairs per year
Eco Products	Redesign products to be biodegradable, organic or low-impact. Example: Patagonia recycled fleece with built-in repairability
Eco Packaging	Replace plastic with cardboard, paper or compostable materials. Example: McDonald's paper straws — 1.8bn plastic straws eliminated
Adv/Disadv	Advantages: reputation, cost savings, legal compliance, investors. Disadvantages: high upfront cost, complex supply chains, greenwashing risk
Greenwashing	False eco-claims. Examples: H&M (EU investigation), VW ($30bn fines, executives imprisoned), Shell (ASA advert ban 2023)